

PhyRes: Physics-Informed Residual Graph Neural Networks for Urban Heat Island Diagnostics

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Abstract—Urban Heat Island (UHI) diagnostics in dense environments require spatial and causal fidelity absent in standard numerical and black-box deep learning models. We present PhyRes, a Physics-Informed Residual Graph Neural Network (GNN) for high-resolution thermal analysis of heterogeneous urban morphologies. Unlike isotropic GNNs, PhyRes enforces hard topological constraints by pruning upwind edges based on instantaneous wind vectors, halving graph complexity while preserving thermodynamic causality. The model integrates 2.5D volumetric building data with ERA5-Land atmospheric reanalysis and satellite indices to construct dynamic, physics-constrained graphs. A dual-head architecture disentangles synoptic weather from localized anthropogenic heat residuals, enabling high-fidelity source apportionment. Evaluated across Bengaluru and Hyderabad, PhyRes achieves sub-degree RMSE values of 0.920°C and 0.610°C, respectively, with building-density correlations reaching up to 0.925 across seasonal test conditions. These results establish PhyRes as a scalable, generalizable diagnostic tool for heat-risk mapping and sustainable urban planning in the Global South.

Keywords—Urban Heat Island, Graph Neural Networks, Physics-Informed Machine Learning, Causal Topology, Advection Pruning, Urban Microclimate, Spatio-Temporal Forecasting

I. INTRODUCTION

A. The Urban Heat Challenge:

Rapid urbanization in the Global South, exemplified by Bengaluru [6], creates heterogeneous microclimates with sub-kilometer thermal variations. Standard meteorological models like ERA5 [2] (31km resolution) lack the morphological fidelity to capture these street-canyon anomalies. While deep learning offers a downscaling path [11], standard CNNs treat urban environments as flat Euclidean grids, ignoring thermodynamic causality (advection and shading). This leads to "over-smoothing" where localized hotspots are mathematically erased [10] and introduces predictive bias by failing to account for wind-driven fluid dynamics.

B. The PhyRes Framework:

To address spatial rigidity and causal entanglement, we introduce the Physics-Informed Residual (PhyRes) framework. PhyRes integrates 2.5D morphological data

[8] and Google Open Buildings footprints and space-borne tropospheric composition soundings [9] to simulate structural shading, thermal mass, and localized combustion proxies. We enforce a Causal Topology where message passing is dynamically pruned along instantaneous wind vectors (**Advection-based Pruning**). This directed graph structure ensures heat flow mirrors physical air currents rather than simple spatial proximity, isolating thermal residuals from the atmospheric baseline with high granularity.

C. Source Apportionment:

Utilizing a dual-head architecture, PhyRes performs source apportionment by disentangling synoptic atmospheric weather (Physics Head) from anthropogenic heat residuals (Residual Head). This provides a high-fidelity diagnostic map of the built environment's direct thermal impact.

II. RELATED WORK

A. Physics vs. Data-Driven Modelling:

Traditional Numerical Weather Prediction (NWP) systems offer physical fidelity but suffer from prohibitive computational overhead [5]. Conversely, satellite-based Land Surface Temperature (LST) retrieval provides spatial coverage but lacks continuous temporal resolution [10]. While LST captures raw thermal signatures, modern diagnostics supplement these observations with high-resolution satellite trace gas sounders like TROPOMI [9] to capture the atmospheric combustion anomalies that serve as proxies for anthropogenic heat flux. Recent landscape frameworks, like those for the Kolkata Metropolitan Area [14], use machine learning to correlate urbanization indices with LST. However, grid-based deep learning (CNNs, GANs) fails in non-Euclidean urban environments where irregular geometries dictate thermal signatures.

B. GNNs and Hard Topological Constraints:

Graph Neural Networks (GNNs) handle irregular topologies but rely on isotropic message passing, causing "Causal Entanglement" where temperature propagates against wind vectors [13]. While Physics-Informed Neural Networks (PINNs) [4] mitigate this via soft loss penalties, they often fail in noisy environments. PhyRes implements **Hard Topological Constraints** by explicitly severing upwind edges [12].

III. METHODOLOGY

A. Study Area and 2.5D Data Integration

Environmental and morphological features were ingested via Google Earth Engine and the Google Open Buildings 2.5D dataset [1]. We transitioned from a 2D grid to a 2.5D spatial matrix by integrating volumetric building heights (H_{avg}) and surface densities (C_{bldg}). The spatio-temporal feature set is summarized in Table 1. We normalized all the values with the help of Z-Score normalization: $x_{norm} = \frac{x - \mu}{\sigma + 10^{-6}}$, where μ and σ are the mean and standard deviation calculated across all nodes and time steps.

Table 1: Spatio-Temporal Feature Set

Feature	Source	Physical Significance
T_{2m}, u_{10m}, v_{10m}	ERA5-Land	Atmospheric baseline and advection vectors.
SWR, STR, LHF	ERA5-Land	Solar/thermal drivers and evaporative flux.
NDVI, NDBI	Landsat 9	Vegetation "pull" and built-up "push" proxies.
H_{avg}, C_{bldg}	Google Open Buildings	Volumetric resistance and thermal storage.
NO_2	Sentinel-5P	Anthropogenic heat/combustion proxy.

B. Causal Graph Construction

1) Advection-based Topological Pruning:

Unlike isotropic GNNs, we enforce thermodynamic causality via a directed masking algorithm [13]. As visualized in Fig. 1, an edge between nodes i and j is preserved only if its orientation aligns with the instantaneous wind vector w_t : $w_t \cdot (pos_j - pos_i) \geq 0$. This eliminates non-causal upwind message passing, halving graph complexity (from 14,936 to 7,441 edges in Bengaluru) while forcing heat flow to follow physical air currents.

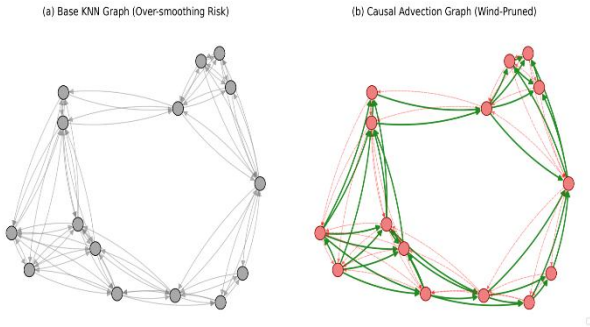


Figure 1: Conceptual schematic of Causal Advection Pruning. (a) A baseline undirected KNN graph allows isotropic message passing, introducing non-physical "upwind" noise and increasing over-smoothing risk. (b) The PhyRes framework enforces a Hard Topological Constraint based on instantaneous wind vectors ($w_t \cdot e_{ij} \geq 0$). Upwind edges are explicitly pruned (dashed red), ensuring heat propagation mirrors physical fluid dynamics.

2) Dynamic Edges

The graph connectivity is modulated on an hourly basis, driven by three physics rules:

- **Solar Shifting:** Attenuates edge weights along solar azimuth vectors to simulate building shadows.
- **Vegetation Evapotranspiration:** Modulates edges at high-NDVI nodes using latent heat flux to capture biomass cooling.
- **Convective Diffusion:** Scales physical distance into effective distance based on wind alignment, making downwind nodes mathematically closer for faster thermal transfer. $d_{ij}^{eff} = \frac{d_{ij}}{1 + K(w_t \cdot e_{ij})}$

3) Static Semantic Edges

- **Feature Similarity (KNN):** K-Nearest Neighbor connections are formed in the morphological feature space (NDBI and Building Height). This allows nodes with similar structural characteristics (e.g., high-rise glass towers) to share learned thermal signatures regardless of spatial distance.
- **Internal Contiguity:** An 8-neighbor adjacency matrix is maintained to represent standard ground-level heat conduction between physically touching land-use parcels.

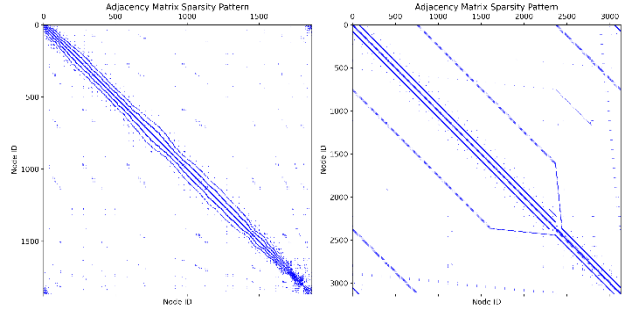


Figure 2: Adjacency matrix sparsity via plt.spy() for (a) Bengaluru and (b) Hyderabad. Diagonal clustering confirms localized GNN message passing.

C. Neural Engine: SA-GAT + LSTM

1) SA-GAT Layer

The architecture utilizes a Spatio-Temporal Graph Attention Network (SA-GAT) [3] to disentangle internal thermal mass from convective advection.

- **Self-Linear Transformation:** Processes the node's internal state via a dedicated linear layer to represent storage.
- **Asymmetric Attention:** The coefficients α_{ii} and α_{ij} are learned via an asymmetric mechanism. This allows the model to identify which specific neighbors act as thermal sources (pushing heat) versus thermal sinks (absorbing/pulling heat).

- **Update Logic:**

$$\mathbf{h}'_i = \sigma \left(\mathbf{a}_{ii} \mathbf{W}_{self} \mathbf{h}_i + \sum_{j \in \mathcal{N}_i} \mathbf{a}_{ij} \mathbf{W}_{neigh} \mathbf{h}_j \right)$$

2) LSTM Backbone: Temporal Thermal Inertia

An LSTM layer [7] processes a 12-hour temporal window to model latent heating and cooling lags. Conditioning predictions at $T+1$ on the cumulative half-day energy balance enables the network to differentiate transient meteorological spikes from sustained structural thermal storage across diverse urban fabrics.

3) Physics-Residual Head Disentanglement

A primary innovation is the dual-output head. The model predicts temperature as the sum of two distinct components: $T_{total} = T_{phys} + T_{res}$

- **T_{phys} (Atmospheric State):** Represents the baseline temperature driven by ERA5 weather physics.
- **T_{res} (Anthropogenic Residual):** Isolates the localized "Urban" component, encapsulating heat emitted from buildings and the cooling effect of parks/water bodies etc.

This explicit architectural decoupling ensures that back-propagation gradients originating from high-frequency urban noise do not contaminate the regularized physics baseline. Consequently, the dual-head framework preserves structural target patterns even when synoptic boundary conditions exhibit high volatility.

D. Physics-Informed Training Strategy:

1) Composite Loss Function

The total loss is the weighted sum of predictive accuracy and physical consistency:

$$L_{total} = L_{MSE} + \alpha L_{phys} + \gamma L_{urban}$$

2) Advection Penalty (L_{phys})

This term penalizes gradients that violate fluid dynamics. If the temperature difference between connected nodes is excessive relative to the wind speed, the penalty increases:

$$L_{phys} = \mathbb{E} \left[\frac{|T_{phys,i} - T_{phys,j}|}{WindSpeed + 1.0} \right]$$

3) **Push-Pull Constraints:** Unlike global penalties, the model uses Spatially-Masked Regularization to enforce hard barriers based on the environment's geometry:

- **Urban Heat Floor (The Push):** Applied where building density > 0.7 . It penalizes any T_{res} falling below 0.8°C , forcing an urban baseline.

$$L_{push} = \text{mean}(\text{ReLU}(0.8 - T_{res}) \cdot \text{Mask}_{urban})$$

- **Vegetation Ceiling (The Pull):** Applied where $\text{NDVI} > 0.6$. It penalizes positive T_{res} , enforcing the cooling effect of transpiring biomass.

$$L_{pull} = \text{mean}(\text{ReLU}(T_{res} - (-0.5)) \cdot \text{Mask}_{veg})$$

These geometric barriers effectively act as a spatial prior, anchoring the residual head to established thermodynamic expectations for high-density and high-vegetation zones to prevent gradient drift in outlier microclimates.

4) Neutral Sparsity: Eliminating "Ghost Heat"

A secondary sparsity loss ($0.005 \cdot T_{res}^2$) is applied exclusively to "Neutral Zones" (low density/low NDVI). This forces the residual toward zero in rural or agricultural buffers, preventing the model from "hallucinating" urban heat in the absence of physical drivers like concrete or traffic.

IV. RESULTS AND DISCUSSION

A. Predictive Accuracy and Seasonal Resilience

The performance of the PhyRes framework was evaluated across two geographically distinct Indian metropolises across summer (April) and winter (December) regimes. Table 2 summarizes the comparative metrics.

In the April Bengaluru development group, PhyRes achieved sub-degree accuracy, effectively mapping the thermal baseline, Fig. 3. In the December test, while the *RMSE* increased to 1.739°C due to winter atmospheric volatility, the Building Correlation ($\text{rBldg}=0.741$) remained high. This confirms that while the absolute temperature magnitude shifted, the model continued to correlate strongly with the urban morphology, correctly identifying thermal hotspots despite increased atmospheric noise.

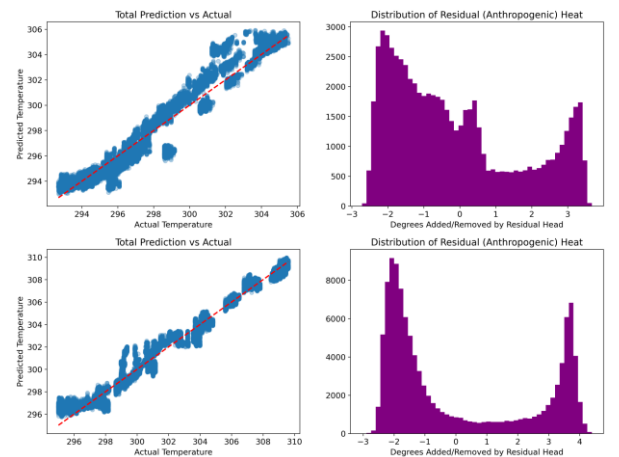


Figure 3: Regression analysis (True vs. Predicted temperature) demonstrates high alignment with the identity line for both (a) Bengaluru (*RMSE* 0.920°C) and (b) Hyderabad (*RMSE* 0.610°C). The tighter convergence in Hyderabad validates the model's successful cross-city generalization, proving internalization of universal thermodynamic urban laws rather than local over-fitting.

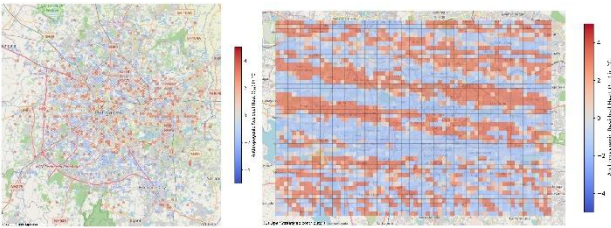
Table 2 : Multi-Scenario Performance & Morphological Fidelity

City	Month	Model	RMSE	rBldg	rVeg
Bengaluru	April	PhyRes	0.920°C	0.800	-0.640
		Baseline	0.732°C	0.002	0.009
		No-Pruning	0.836°C	-0.019	0.091
	Dec	PhyRes	1.739°C	0.741	-0.702
		Baseline	1.310°C	-0.006	0.008
		No-Pruning	1.603°C	0.018	-0.010
Hyderabad	April	PhyRes	0.610°C	0.904	-0.483
		Baseline	0.615°C	0.007	-0.004
		No-Pruning	0.951°C	0.067	-0.008
	Dec	PhyRes	0.764°C	0.925	-0.612
		Baseline	1.151°C	0.005	-0.006
		No-Pruning	0.855°C	0.000	0.006

Note: rBldg and rVeg denote the Pearson correlation of the Residual Heat T_{res} with Building Density and NDVI, respectively. Baseline models (MSE-only) yield 0.00 correlation as they lack a residual disentanglement head.

B. Cross-City Generalization

To evaluate the portability of the framework, the architectural configuration and hyperparameter layout optimized for Bengaluru were utilized to initialize and train a separate model instance from scratch on the Hyderabad spatial topology, as shown in Fig. 4. The model outperformed its Bengaluru baseline, achieving an *RMSE* of **0.610°C** and a notably high building correlation of **0.904**. This suggests that the PhyRes architecture is not merely memorizing local noise but has successfully internalized the universal thermodynamic relationship between 2.5D building volume and heat retention.



lies across distinct city structures and contrasting seasonal regimes.

C. Ablation and Structural Health

The necessity of the dual-head architecture and Hard Topological Constraints is proven through ablation testing against a Vanilla Baseline (single-head) and a No-Pruning (static graph) model. Standard single-head GNNs suffered from severe over-smoothing, where distinct thermal signatures blurred into a global mean. By explicitly decoupling the synoptic atmospheric baseline from the localized anthropogenic residual, the dual-head architecture successfully mitigated this node collapse, preserving the high-frequency signatures required for micro-climate diagnostics. While the dual-head design

acts as the primary defense against over-smoothing, the wind-driven advection pruning serves as a critical topological constraint. It reduces the spatial message-passing complexity by 50.18% (from 14,936 to 7,441 active edges in Bengaluru) while strictly enforcing fluid-dynamic causality. By ensuring structural geometry overrides synoptic cooling without introducing non-physical upwind noise, the framework provides a high-fidelity basis for targeted mitigation strategies, such as cool-roof mandates and optimized green infrastructure deployment in high-residual industrial corridors.

V. CONCLUSION

This paper introduced PhyRes, a Physics-Informed Residual GNN framework that integrates 2.5D volumetric data to disentangle morphological urban heat from synoptic atmospheric states. By transitioning from stochastic "soft" penalties to deterministic "hard" topological constraints via wind-driven causal pruning, the architecture achieves sub-degree microclimate forecasting across Bengaluru and Hyderabad. The dual-head mechanism isolates localized anthropogenic residuals from regional weather trends, acting as an actionable diagnostic tool for urban heat analysis. Furthermore, successful cross-city architectural replication demonstrates that the underlying PhyRes framework successfully codifies generalized structural thermodynamic principles rather than overfitting to city-specific geographic noise, establishing a scalable paradigm for climate risk mapping in the Global South. Future work will prioritize integrating dynamic traffic flux metrics and 3D street-canyon radiative transfer modeling to further refine the diagnostic resolution of the anthropogenic residual.

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DATA AVAILABILITY STATEMENT

The source code for the PhyRes framework, pre-processed 2.5D urban datasets, and training logs are hosted at <https://github.com/AyushG-1210/PhyRes-Physics-Informed-UHI-Diagnostics> for academic reproducibility and further research.

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